

consider R.H.S = $(0+1)^*$
 $= \{ \epsilon, 0, 00, 1, 11, 01, 10, \dots \}$
 $= \{ \text{any combination of 0's any combination of 1's, any combination of 0's \& 1's} \}$

hence L.H.S = R.H.S is proved

Example - 2

show that $(ab)^* \neq (a^*b^*)$

Solⁿ $\rightarrow$ Consider L.H.S = $(ab)^*$
 $= \{ \epsilon, ab, abab, ababab, \dots \}$

Consider R.H.S = (a^*b^*)
 $= \{ \epsilon, a, aa, b, bb, bbb, ab, aabb, \dots \}$

Note in L.H.S, regular expression gives the string of 'ab' 'abab' so on. But in R.H.S. there is no possibility of abab.

Thus L.H.S $\neq$ R.H.S.

* Properties of Regular languages

- Closure properties of regular languages

If certain languages are regular and L is formed from them by certain operation then L is also regular. These properties are called closure properties of regular languages. Such languages represents the same class of regular which is closed under certain specific operations.

The closure properties of regular languages are given below.

1. Union of two regular languages is Regular.
2. The intersection of two regular language is regular.
3. Complement of regular language is regular.
4. The difference of two regular language is regular.
5. The reversal of regular language is regular.
6. Closure operation on regular language is Regular.
7. Concatenation of regular language is regular.
8. A homomorphism of regular language is regular.
9. The inverse homomorphism of regular language is regular.



* Decision properties of regular languages.
following is the set of questions for determining the decision properties of regular language.

1. Is a particular string N belongs to some language L ?

Let L be a language which can be represent by DFA. If string w is chosen for simulating this DFA processing and then if w reaches to an accept state, starting from the start state.

2. Is the language empty?

The regular language is modeled by finite automata to decide whether it is empty or not. If there exist some path from start state to final state for deriving such language then the given language is not empty.

3. Do the two description of a languages represent the same language.

Any regular language can be representative by NFA, DFA, or regular expression.

Thus representation can be interconverted. This property is called equivalence of

corresponding two models.

* Pumping Lemma for Regular language

It is used to prove that, language is not regular.

* every finite language is regular.

* If we are able to design a FA for any language then language is regular.

* By Using Pumping Lemma:

If we are able to find some pattern in language then may be that language is regular but if we don't get any pattern then definitely language is not regular.

- Pumping lemma states that,

If L is regular set, then there exist a constant n , such that for every word z in L and $|z| \geq n$, we can write $z = uvw$ in such a way that $|uv| \leq n$, $|v| \geq 1$, and for all $i \geq 0$, $uv^i w$ is in L .

Example -

Prove that $L = \{0^i 2^j / i \geq 1\}$ is non-regular.

Proof -

Let n be a constant of pumping lemma.
select $z = 0^{n^2}$ this ensure that z is in L , $|z| \geq n$.
if we write $z = uvw$ then the possible chances
of satisfying the conditions.

$1 \leq |v| \leq n$ & $|uv| \leq n$ are
 $v = 0^p$; where $1 \leq p \leq n$ i.e. v can be chosen to
be a string of minimum one zero & maximum
of n 0's.

for this choice of v , $|uv^2w|$ (i.e. the length
of uv^2w) will be between n^2+1 & n^2+n
since $n^2+1 > n^2$ & $n^2+n < (n+1)^2$. we
conclude that length of uv^2w is not
a perfect square of any integer,
hence uv^2w can not belong to L .

$\therefore$ we get contradiction to pumping lemma.
hence proving that the given language
is not regular.

* Applications of Regular Expression.

Regular Expression in Unix

There are various unix notation's used for regular expression. these notation's have many additional capabilities & feature. this notations, have ability to name and refer previous strings that have matched pattern. this ability helps in recognising non regular language.

e.g. symbol $\cdot$ stands for single character. sequence $[1, 2, 3, \dots, 10]$ means $1+2+3+\dots+10$. (n) means n copies of

Lexical Analyzer's

Lexical analyzer's used in compiler's. that's of lexical analyze is to scan out the input program & separate out the token's.

Example - Identifier is a category of token in the source language & it can be identified by regular expression.

finding patterns in text.

Regular expressions are useful notations. used for finding the patterns from given text. a large class of pattern can be identified by



regular expression.

* Question Bank

- ① Discuss Application of FA & regular expression. [4]
- ② Using pumping lemma, prove that $\{0^i 2^0 / i \text{ is an integer, } i \geq 1\}$ is not regular. [6]
- ③ prove formula [4]
$$(x^* s^*)^* = (x + s)^*$$
$$(a b)^* \neq a^* b^*$$
- ④ Construct a DFA with reduced state equivalent to $10 + (0 + 11)0^* 1$ [6]
- ⑤ Define Pumping lemma and apply it to prove following [6]
$$L = \{0^m 1^n 0^{m+n} / m \geq 1 \text{ \& } n \geq 1\}$$
 is not regular.